

§ HP filter: decompose economic series into trend & cyclical component

$$y_t = y_t^c + y_t^T$$

① Penalty of calling everything trend

$$\min_{\{y_t^T\}_{t=1}^N} \mathcal{L} = \sum_{t=1}^N (y_t - y_t^T)^2 + \lambda \sum_{t=2}^{N-1} [(y_{t+1}^T - y_t^T) - (y_t^T - y_{t-1}^T)]^2$$

② change in trend growth rate, penalty of wiggles.

If  $\lambda$  huge close to straight line in logs

If  $\lambda=0$ , wiggle a lot. fit to  $y_t$ .

$$\text{Define } \Delta^2 y_t^T = y_{t+1}^T - 2y_t^T + y_{t-1}^T \text{ second order difference}$$

$$\frac{\partial \mathcal{L}}{\partial y_t^T} = 0 \Rightarrow \frac{\partial \mathcal{L}}{\partial y_t^T} = 2(y_t^T - y_t)$$

$$\text{Since } \Delta^2 y_{t-1}^T = y_t^T - 2y_{t-1}^T + y_{t-2}^T \text{ (coefficient +1)}$$

$$\Delta^2 y_t^T = y_{t+1}^T - 2y_t^T + y_{t-1}^T \text{ (coefficient -2)}$$

$$\Delta^2 y_{t+1}^T = y_{t+2}^T - 2y_{t+1}^T + y_t^T \text{ (coefficient +1)}$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial y_t^T} \sum_{s=2}^{N-1} (\Delta^2 y_s^T)^2 &= 2 \Delta^2 y_{t-1}^T \frac{\partial \Delta^2 y_{t-1}^T}{\partial y_t^T} + 2 \Delta^2 y_t^T \frac{\partial \Delta^2 y_t^T}{\partial y_t^T} + 2 \Delta^2 y_{t+1}^T \frac{\partial \Delta^2 y_{t+1}^T}{\partial y_t^T} \\ &= 2 (\Delta^2 y_{t-1}^T - 2 \Delta^2 y_t^T + \Delta^2 y_{t+1}^T) \end{aligned}$$

$$\text{Since } \Delta^2 y_{t-1}^T = y_t^T - 2y_{t-1}^T + y_{t-2}^T$$

$$-2 \Delta^2 y_t^T = -2(y_{t+1}^T - 2y_t^T + y_{t-1}^T) = -2y_{t+1}^T + 4y_t^T - 2y_{t-1}^T$$

$$\Delta^2 y_{t+1}^T = y_{t+2}^T - 2y_{t+1}^T + y_t^T$$

$$\frac{\partial \mathcal{L}}{\partial y_t^T} = \frac{\partial \mathcal{L}}{\partial y_t^T} + \lambda \frac{\partial}{\partial y_t^T} (\Delta^2 y_s^T)^2 = 0 \Rightarrow (1+6\lambda)y_t^T - 4\lambda y_{t-1}^T - 4\lambda y_{t+1}^T + \lambda y_{t-2}^T + \lambda y_{t+2}^T = y_t$$

$3 \leq t \leq N-2$

$$\text{At } t=1: (1+\lambda)y_1^T - 2\lambda y_2^T + \lambda y_3^T = y_1$$

(only  $\Delta^2 y_{t+1}^T$  exists)

$$\text{At } t=2: -2\lambda y_1^T + (1+5\lambda)y_2^T - 4\lambda y_3^T + \lambda y_4^T = y_2$$

(only  $\Delta^2 y_{t-1}^T$  &  $\Delta^2 y_t^T$  exist)

Symmetrically for  $t=N$  and  $t=N-1$ , collect  $\{y_t\}$  into  $N \times 1$  vector  $y$ .  $\{y_t^T\}$  into  $y^T$ , then we have  $A \cdot y^T = y \Rightarrow y^T = A^{-1}y \Rightarrow y^c = y - A^{-1}y$ .

$$\text{Where, } A = \begin{pmatrix} 1+\lambda & -2\lambda & \lambda & 0 & 0 & \dots & 0 & 0 & 0 \\ -2\lambda & 1+5\lambda & -4\lambda & \lambda & 0 & \dots & 0 & 0 & 0 \\ \lambda & -4\lambda & 1+6\lambda & -4\lambda & \lambda & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1+6\lambda & -4\lambda & \lambda & 0 & 0 \\ 0 & 0 & 0 & \dots & -4\lambda & 1+5\lambda & -2\lambda & 0 & 0 \\ 0 & 0 & 0 & \dots & \lambda & -2\lambda & 1+\lambda & 0 & 0 \end{pmatrix}$$



Let  $D$  be the  $(N-2) \times N$  second difference matrix, whose  $t$ -th row ( $t=2 \dots N-1$ )

looks like:  $[0 \dots 1 \ -2 \ 1 \dots 0]$

so that  $(Dy^T)_t = \Delta^2 y_t^T$

HP objective becomes:  $\min_{y^T} (y - y^T)'(y - y^T) + \lambda (Dy^T)'(Dy^T)$

F.O.C.  $-2(y - y^T) + 2\lambda D'Dy^T = 0 \Rightarrow (I + \lambda D'D)y^T = y$ .

$\Rightarrow A = I + \lambda \boxed{D'D}$  5-diagonal pattern

interior diagonal:  $[0 \dots 1 \ -2 \ 1 \dots 0]$   $\begin{bmatrix} 0 \\ \vdots \\ 1 \\ -2 \\ 1 \\ \vdots \\ 0 \end{bmatrix} = 6$

first-off diagonal:  $[0 \dots 1 \ -2 \ 1 \ 0 \dots 0]$   $\begin{bmatrix} 0 \\ \vdots \\ 0 \\ -1 \\ 1 \\ \vdots \\ 0 \end{bmatrix} = -4$

second-off diagonal:  $[0 \dots 1 \ -2 \ 1 \ 0 \ 0 \dots 0]$   $\begin{bmatrix} 0 \\ \vdots \\ 0 \\ 0 \\ -1 \\ 1 \\ \vdots \\ 0 \end{bmatrix} = 1$

$\lambda \uparrow \Rightarrow$  The smoother the series  $y_t^T$

Pros: ① Simple & transparent. (defined by one objective)

② Produces a smooth trend with tunable smoothness  $\lambda$

③ Widely used in Macro

Cons: ① End-point problem: start and end of sample is distorted.

②  $\lambda$  sensitivity: moments (volatility, correlations) change a lot with  $\lambda$ .

③ Spurious dynamics: HP can induce artificial serial correlations and distort cross-corr between series (create cycles)

④ Leakage: Low frequency cycles leak into trend.

⑤ Not structural: no model interpretation

Why  $\lambda=1600$  is the quarterly standard?

Rule of thumb:  $\lambda = 100 \times (\text{periods per year})^2$ ,  $\lambda_{\text{quarter}} = 100 \times 4^2 = 1600$   
 $\lambda_{\text{month}} = 100 \times 12^2 = 14400$   
 $\lambda_{\text{annual}} = 100 \times 1^2 = 100$



# 6 Long and Plosser (1983, JPE)

Setup:  $U = \max_{C_t, l_t, k_t} E_0 \left[ \sum_{t=0}^{\infty} \beta^t [\theta \ln C_t + (1-\theta) \ln L_t] \right]$

s.t.  $k_t = z_t k_{t-1}^\alpha N_t^{1-\alpha} - C_t$  and  $k_0$  is given. Resource constraint ( $C_t + k_t = Y_t$ )

$N_t + L_t = 1$  *L: leisure N: working hour. Total hour normalized to 1.*

where  $\log z_t = (1-\psi) \log \bar{z} + \psi \log z_{t-1} + \varepsilon_t$ ,  $\varepsilon_t \stackrel{i.i.d}{\sim} N(0, \sigma^2)$ ,  $\psi < 1$

Note: capital completely turns over,  $(1-\delta)k_{t+1} = 0$   $\delta = 1$

Lagrangian:  $\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t [\theta \ln C_t + (1-\theta) \ln (1-N_t)] + \lambda_t (z_t k_{t-1}^\alpha N_t^{1-\alpha} - C_t - k_t)$

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \Rightarrow \frac{\theta}{C_t} = \lambda_t \quad (1)$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = 0 \Rightarrow -\frac{1-\theta}{1-N_t} + \lambda_t (1-\alpha) z_t k_{t-1}^\alpha N_t^{-\alpha} = 0 \Rightarrow \frac{1-\theta}{1-N_t} = (1-\alpha) \lambda_t z_t k_{t-1}^\alpha N_t^{-\alpha} \quad (2)$$

*labor-leisure condition (intra-temporal)*

$$\frac{\partial \mathcal{L}}{\partial k_t} = 0 \Rightarrow -\lambda_t \beta^t + \beta^{t+1} E_t \left[ \lambda_{t+1} \frac{\partial}{\partial k_t} (z_{t+1} k_t^\alpha N_{t+1}^{1-\alpha}) \right] = 0.$$

$$\Rightarrow \lambda_t = \beta E_t (\lambda_{t+1} \alpha z_{t+1} k_t^{\alpha-1} N_{t+1}^{1-\alpha}) \quad (3) \quad \dots \text{Euler equation (intertemporal)}$$

TVC:  $\lim_{T \rightarrow \infty} \beta^T U_C(C, L) k_{T+1} = 0 \quad (4)$

Closed-form solution; constant savings rate & constant labor

\* Tractability: log utility + Cobb-Douglas + full depreciation  $\Rightarrow$  constant savings rate/hours.

$$Y_{t+1} = z_{t+1} k_t^\alpha N_{t+1}^{1-\alpha} \Rightarrow \alpha z_{t+1} k_t^{\alpha-1} N_{t+1}^{1-\alpha} = \alpha \frac{Y_{t+1}}{k_t}$$

Plug into (3)  $\Rightarrow \lambda_t = \beta E_t (\lambda_{t+1} \alpha \frac{Y_{t+1}}{k_t})$  plug in (1)  $\Rightarrow$

$$\frac{\theta}{C_t} = \beta E_t \left( \frac{\theta}{C_{t+1}} \alpha \frac{Y_{t+1}}{k_t} \right) \Rightarrow \frac{1}{C_t} = \beta \alpha E_t \left[ \frac{Y_{t+1}}{C_{t+1}} \frac{1}{k_t} \right] \quad (5)$$

Guess and verify:  $C_t = (1-s)Y_t$ ,  $s$  is the savings rate,  $k_t = sY_t$

$$\Rightarrow \frac{Y_{t+1}}{C_{t+1}} = \frac{1}{1-s} \Big|_{(5)} \Rightarrow \frac{1}{(1-s)Y_t} = \alpha \beta \left( \frac{1}{1-s} \right) \frac{1}{sY_t} \Rightarrow s = \alpha \beta$$

$$\Rightarrow C_t = (1-\alpha\beta)Y_t \quad k_t = \alpha\beta Y_t \quad \left\{ \begin{array}{l} \alpha \uparrow, \text{ capital income share } \uparrow \Rightarrow \text{savings rate } \uparrow \\ \beta \uparrow, \text{ discount factor } \downarrow \Rightarrow \text{savings rate } \uparrow \end{array} \right.$$

Plug into (2)  $\Rightarrow \frac{1-\theta}{1-N_t} = (1-\alpha) \frac{\theta}{C_t} \frac{Y_t}{N_t} = (1-\alpha) \frac{\theta}{(1-\alpha\beta)Y_t} \frac{Y_t}{N_t}$

$$\Rightarrow (1-\theta)(1-\alpha\beta)N_t = (1-\alpha)\theta(1-N_t) \Rightarrow N_t = \bar{N} = \frac{\theta(1-\alpha)}{\theta(1-\alpha) + (1-\theta)(1-\alpha\beta)}$$



Since  $N_t = \text{constant } \forall t \Rightarrow \hat{n}_t = 0$

(34)

$$K_t = \alpha \beta Y_t \quad C_t = (1 - \alpha \beta) Y_t \Rightarrow \hat{k}_t = \hat{c}_t = \hat{y}_t \quad \Bigg| \Rightarrow \hat{k}_t = \hat{c}_t = \hat{y}_t = \hat{z}_t + \alpha \hat{k}_{t-1}$$

$$\hat{k}_t = \hat{z}_t + \alpha \hat{k}_{t-1} + (1 - \alpha) \hat{n}_t = \hat{z}_t + \alpha \hat{k}_{t-1}$$

$$\hat{z}_t = \psi \hat{z}_{t-1} + \varepsilon_t$$

TFP shock alone can generate output / consumption / investment fluctuations, but ① labor does not fluctuate at all, and ②  $C, k, y$  moves one-for-one.

Consumption is usually smoother than output, capital is usually more volatile

③ Labor productivity moves like output in model.  $(\frac{\hat{Y}}{\hat{N}})_t = \hat{y}_t - \hat{n}_t = \hat{y}_t$ , while in data.

$$\bar{\sigma}(\frac{Y}{N}) \ll \bar{\sigma}(Y) \quad \& \quad \text{Corr}(\frac{Y}{N}, Y) = 0.55$$

④ Prices (MRS, MRT, MPL, MPK) too tight with output & TFP, hard to generate weak wage cyclicality & interest rate counter-cyclicality.

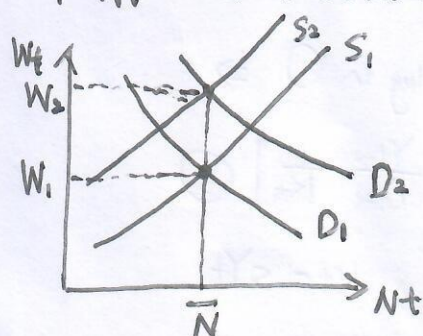
Labor market equilibrium:

Household's problem:  $U_{N,t} + \lambda_t W_t = 0 \Rightarrow U_{N,t} + V_{C,t} W_t = 0 \dots$  Labor supply

Firm's problem:  $\max_{K_t, N_t} Y_t - W_t N_t - R_t K_{t+1} \dots$  Labor demand

$$\Rightarrow W_t = Z_t F_{N,t} = Z_t \frac{Y_t}{N_t} (1 - \alpha)$$

$$\Rightarrow \begin{cases} W_t = \frac{1-\theta}{\theta} \frac{C_t}{1-N_t} & \Rightarrow W_t = \frac{1-\theta}{\theta} \frac{(1-\alpha\beta) Z_t k_{t-1}^\alpha N_t^{1-\alpha}}{1-N_t} \dots Z_t \uparrow k_{t-1}^\alpha \text{ shifts labor supply up by same factor } Z_t k_{t-1}^\alpha \\ W_t = (1-\alpha) Z_t k_{t-1}^\alpha N_t^{-\alpha} \dots \sim \text{demand up} \sim \end{cases}$$



labor supply's wealth effect exactly offset by labor demand increase

9 Hansen (1985) Basic

Expand log preference to make labor respond to shocks.

Household's problem:  $\max_{\{C_t, L_t, N_t\}} E_0 \sum_{t=0}^{\infty} \beta^t u(C_t, L_t) \equiv E_0 \sum_{t=0}^{\infty} \beta^t u(C_t, 1 - N_t)$

$$\text{s.t.} \quad N_t + L_t = 1$$

$$C_t + K_t = W_t N_t + R_t K_{t-1} + (1 - \delta) K_{t-1} \dots \text{budget constraint}$$

$$* K_t - (1 - \delta) K_{t-1} = I_t \dots \text{investment}$$



$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t [u(C_t, 1-N_t) + \lambda_t (W_t N_t + R_t K_{t+1} + (1-\delta)k_{t+1} - C_t - k_t)] \quad (35)$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \Rightarrow U_{C,t} = \lambda_t \quad \text{marginal utility of consumption}$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = 0 \Rightarrow U_{N,t} + \lambda_t W_t = 0 \Rightarrow -U_{N,t} = \lambda_t W_t \rightarrow \text{marginal utility value of one more unit of wage income}$$

$$\Rightarrow -U_{N,t} = U_{C,t} W_t$$

$$\frac{\partial \mathcal{L}}{\partial k_t} = 0 \Rightarrow -\lambda_t \beta^t + \beta^t E_t [\lambda_{t+1} R_{t+1} + \lambda_{t+1} (1-\delta)] = 0$$

$$\Rightarrow U_{C,t} = \beta E_t [U_{C,t+1} (R_{t+1} + 1-\delta)]$$

ii. Firm's problem:  $\max_{N_t, K_{t+1}} \Pi_t = Y_t - W_t N_t - R_t K_{t+1}$

s.t.  $Y_t = Z_t K_{t+1}^\alpha N_t^{1-\alpha}$ ,  $\log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t+1} + \varepsilon_t$

$$\Rightarrow \frac{\partial \Pi_t}{\partial K_{t+1}} = 0 \Rightarrow \alpha Z_t K_{t+1}^{\alpha-1} N_t^{1-\alpha} = R_t \Rightarrow \alpha \frac{Y_t}{K_{t+1}} = R_t$$

$$\frac{\partial \Pi_t}{\partial N_t} = 0 \Rightarrow (1-\alpha) Z_t K_{t+1}^\alpha N_t^{-\alpha} = W_t \Rightarrow (1-\alpha) \frac{Y_t}{N_t} = W_t$$

Let utility function follows,  $U(C_t, 1-N_t) = \ln C_t + A \ln(1-N_t)$

$$U_{C,t} = \frac{1}{C_t} \quad U_{N,t} = -\frac{A}{1-N_t}$$

$$\Rightarrow \left\{ \begin{array}{l} 1 = \beta E_t \left[ \frac{C_t}{C_{t+1}} (R_{t+1} + 1-\delta) \right] \\ \frac{A}{1-N_t} = \frac{1}{C_t} W_t \\ W_t = (1-\alpha) Z_t K_{t+1}^\alpha N_t^{-\alpha} \\ R_t = \alpha Z_t K_{t+1}^{\alpha-1} N_t^{1-\alpha} \\ \Pi_t = Z_t K_{t+1}^\alpha N_t^{1-\alpha} - W_t N_t - R_t K_{t+1} = 0 \\ C_t + k_t = W_t N_t + R_t K_{t+1} + (1-\delta)k_{t+1} \\ \log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t+1} + \varepsilon_t \end{array} \right. \Rightarrow A C_t = (1-N_t) (1-\alpha) Z_t K_{t+1}^\alpha N_t^{-\alpha}$$

Not substitute out  $R_t, W_t$  to make a system of equations of  $(C_t, N_t, K_t, Z_t)$

$$\Rightarrow \left\{ \begin{array}{l} 1 = \beta E_t \left[ \frac{C_t}{C_{t+1}} (\alpha Z_{t+1} K_{t+1}^{\alpha-1} N_{t+1}^{1-\alpha} + 1-\delta) \right] \\ A C_t = (1-N_t) (1-\alpha) Z_t K_{t+1}^\alpha N_t^{-\alpha} \\ K_t = Z_t K_{t+1}^\alpha N_t^{1-\alpha} + (1-\delta) K_{t+1} - C_t \\ \log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t+1} + \varepsilon_t \end{array} \right.$$



At Steady state:  $K_t = K_{t+1} = \bar{K}$   $C_t = C_{t+1} = \bar{C}$   $N_t = N_{t+1} = \bar{N}$

(36)

$$\bar{K} = \bar{z} \bar{K}^\alpha \bar{N}^{1-\alpha} + (1-\delta)\bar{K} - \bar{C} \Rightarrow \bar{C} = \bar{z} \bar{K}^\alpha \bar{N}^{1-\alpha} - \delta \bar{K} \quad (1)$$

$$1 = \alpha\beta \bar{z} \bar{K}^{\alpha-1} \bar{N}^{1-\alpha} + (1-\delta)\beta \quad (2)$$

$$A\bar{C} = (1-\bar{N})(1-\alpha)\bar{z} \bar{K}^\alpha \bar{N}^{-\alpha} \quad (3) \Rightarrow \bar{z} \bar{K}^\alpha \bar{N}^{1-\alpha} - \delta \bar{K} = \frac{1}{A} ((1-\alpha)\bar{z} \bar{K}^\alpha \bar{N}^{-\alpha} - (1-\alpha)\bar{z} \bar{K}^\alpha \bar{N}^{1-\alpha})$$

$$\Rightarrow \bar{z} \bar{K}^{\alpha-1} \bar{N}^{1-\alpha} + \frac{1-\alpha}{A} \bar{z} \bar{K}^{\alpha-1} \bar{N}^{1-\alpha} = \frac{1-\alpha}{A} \bar{z} \bar{K}^{\alpha-1} \bar{N}^{-\alpha} + \delta$$

$$\Rightarrow \bar{z} \bar{K}^{\alpha-1} \bar{N}^{1-\alpha} = \frac{A}{1-\alpha+A} \left[ \frac{1-\alpha}{A} \bar{z} \bar{K}^{\alpha-1} \bar{N}^{-\alpha} + \delta \right]$$

plug in (2)  $\Rightarrow 1 = \frac{A\alpha\beta}{1-\alpha+A} \left[ \frac{1-\alpha}{A} \bar{z} \bar{K}^{\alpha-1} \bar{N}^{-\alpha} + \delta \right] + (1-\delta)\beta$

$$\Rightarrow \bar{z} \bar{K}^{\alpha-1} \bar{N}^{-\alpha} = \left[ \frac{1 - (1-\delta)\beta}{A\alpha\beta} (1-\alpha+A) - \delta \right] \frac{A}{1-\alpha}$$

$$\bar{z} \bar{K}^{\alpha-1} \bar{N}^{1-\alpha} = \frac{1 - (1-\delta)\beta}{\alpha\beta}$$

$$\Rightarrow \bar{N} = \left[ \frac{(1 - (1-\delta)\beta)(1-\alpha+A) - A\delta\alpha\beta}{A\alpha\beta} \right]^{-1} \cdot \frac{(1 - (1-\delta)\beta)(1-\alpha)}{A\alpha\beta}$$

$$\Rightarrow \bar{N} = \frac{A\alpha\beta}{(1 - (1-\delta)\beta)(1-\alpha+A) - A\delta\alpha\beta} \cdot \frac{(1 - (1-\delta)\beta)(1-\alpha)}{A\alpha\beta} = \left[ \frac{(1 - (1-\delta)\beta)(1-\alpha+A) - A\delta\alpha\beta}{(1 - (1-\delta)\beta)(1-\alpha)} \right]^{-1}$$

$$= \left[ \frac{1-\alpha+A}{1-\alpha} - \frac{A\delta\alpha\beta}{(1 - (1-\delta)\beta)(1-\alpha)} \right]^{-1} = \left[ 1 + \frac{A}{1-\alpha} - \frac{A}{1-\alpha} \frac{\delta\alpha\beta}{1 - (1-\delta)\beta} \right]^{-1}$$

$$= \left[ 1 + \frac{A}{1-\alpha} \left[ 1 - \frac{\alpha\beta\delta}{1 - (1-\delta)\beta} \right] \right]^{-1}$$

$$\bar{K} = \left[ \frac{1 - (1-\delta)\beta}{\bar{z}\alpha\beta} \bar{N}^{\alpha-1} \right]^{\frac{1}{\alpha-1}} = \bar{N} \left[ \frac{\bar{z}\alpha}{\beta^{1-(1-\delta)\beta}} \right]^{\frac{1}{1-\alpha}}$$

$$\bar{C} = \bar{z} \bar{K}^\alpha \bar{N}^{1-\alpha} - \delta \bar{K} = \bar{N}^\alpha \left[ \frac{\bar{z}\alpha}{\beta^{1-(1-\delta)\beta}} \right]^{\frac{\alpha}{1-\alpha}} \bar{N}^{1-\alpha} \bar{z} - \delta \bar{K}$$

$$= \bar{z}^{\frac{1}{1-\alpha}} \bar{N} \left[ \frac{\alpha}{\beta^{1-(1-\delta)\beta}} \right]^{\frac{\alpha}{1-\alpha}} - \delta \bar{N} \bar{z}^{\frac{1}{1-\alpha}} \left[ \frac{\alpha}{\beta^{1-(1-\delta)\beta}} \right]^{\frac{1}{1-\alpha}}$$

$$= \bar{N} \bar{z}^{\frac{1}{1-\alpha}} \left[ \left( \frac{\alpha}{\beta^{1-(1-\delta)\beta}} \right)^{\frac{\alpha}{1-\alpha}} - \delta \cdot \left( \frac{\alpha}{\beta^{1-(1-\delta)\beta}} \right)^{\frac{1}{1-\alpha}} \right]$$

Derive the log-linearized system:

$$(i) \log z_t = (1-\psi) \log \bar{z} + \psi \log z_{t+1} + \varepsilon_t \Rightarrow \log \bar{z} e^{\hat{z}_t} = (1-\psi) \log \bar{z} + \psi \log \bar{z} e^{\hat{z}_{t+1}} + \varepsilon_t$$

$$\Rightarrow \hat{z}_t + \log \bar{z} = (1-\psi) \log \bar{z} + \psi \log \bar{z} + \psi \hat{z}_{t+1} + \varepsilon_t \Rightarrow \hat{z}_t = \psi \hat{z}_{t+1} + \varepsilon_t$$



$$(ii) K_t = z_t k_{t-1}^\alpha N_t^{1-\alpha} + (1-\delta) k_{t-1} - c_t$$

$$K(C + \hat{k}_t) = \bar{z}(C + \hat{z}_t) \bar{K}^\alpha (C + \hat{k}_{t-1})^{1-\alpha} \bar{N}^{1-\alpha} (C + \hat{n}_t)^{1-\alpha} + (1-\delta) \bar{K} (C + \hat{k}_{t-1}) - \bar{C} (C + \hat{c}_t)$$

$$\Rightarrow \bar{K} + \bar{K} \hat{k}_t = \bar{z} \bar{K}^\alpha \bar{N}^{1-\alpha} (C + \hat{z}_t) (C + \alpha \hat{k}_{t-1} + (1-\alpha) \hat{n}_t) + (1-\delta) \bar{K} + (1-\delta) \bar{K} \hat{k}_{t-1} - \bar{C} - \bar{C} \hat{c}_t$$

$$\Rightarrow \bar{K} + \bar{K} \hat{k}_t = \bar{z} \bar{K}^\alpha \bar{N}^{1-\alpha} \cdot C + \alpha \hat{k}_{t-1} + \hat{z}_t + \alpha \hat{k}_{t-1} \hat{z}_t + (1-\alpha) \hat{n}_t + \alpha (1-\alpha) \hat{k}_{t-1} \hat{n}_t + (1-\alpha) \hat{n}_t \hat{z}_t + \alpha (1-\alpha) \hat{k}_{t-1} \hat{n}_t \hat{z}_t + (1-\delta) \bar{K} + (1-\delta) \bar{K} \hat{k}_{t-1} - \bar{C} \hat{c}_t$$

$$\Rightarrow \bar{K} \hat{k}_t = \alpha \bar{Y} \hat{k}_{t-1} + \bar{Y} \hat{z}_t + (1-\alpha) \bar{Y} \hat{n}_t + (1-\delta) \bar{K} \hat{k}_{t-1} - \bar{C} \hat{c}_t$$

$$\Rightarrow \hat{k}_t = \alpha \frac{\bar{Y}}{\bar{K}} \hat{k}_{t-1} + \frac{\bar{Y}}{\bar{K}} \hat{z}_t + (1-\alpha) \frac{\bar{Y}}{\bar{K}} \hat{n}_t + (1-\delta) \hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}} \hat{c}_t$$

$$\Rightarrow \hat{k}_t = \left( \alpha \frac{\bar{Y}}{\bar{K}} + 1 - \delta \right) \hat{k}_{t-1} + (1-\alpha) \frac{\bar{Y}}{\bar{K}} \hat{n}_t + \frac{\bar{Y}}{\bar{K}} \hat{z}_t - \frac{\bar{C}}{\bar{K}} \hat{c}_t$$

$$(iii) A c_t = (1 - N_t) (1-\alpha) z_t k_{t-1}^\alpha N_t^{-\alpha}$$

$$\Rightarrow A \bar{C} (C + \hat{c}_t) = [1 - \bar{N} (C + \hat{n}_t)] (1-\alpha) \bar{z} (C + \hat{z}_t) \bar{K}^\alpha (C + \alpha \hat{k}_{t-1}) \bar{N}^{-\alpha} (1 - \alpha \hat{n}_t)$$

$$\Rightarrow A \bar{C} + A \bar{C} \hat{c}_t = (1 - \bar{N}) (1-\alpha) \bar{z} \bar{K}^\alpha \bar{N}^{-\alpha} (C + \hat{z}_t) (C + \alpha \hat{k}_{t-1}) (1 - \alpha \hat{n}_t)$$

$$- \bar{N} \hat{n}_t (1-\alpha) \bar{z} (C + \hat{z}_t) \bar{K}^\alpha (C + \alpha \hat{k}_{t-1}) \bar{N}^{-\alpha} (1 - \alpha \hat{n}_t)$$

since  $\hat{n}_t \hat{z}_t = \hat{n}_t \hat{k}_{t-1} = \hat{n}_t^2 \dots = 0$  only has 1 term

$$\Rightarrow A \bar{C} + A \bar{C} \hat{c}_t = (1 - \bar{N}) (1-\alpha) \bar{z} \bar{K}^\alpha \bar{N}^{-\alpha} C + \alpha \hat{k}_{t-1} + \hat{z}_t - \alpha \hat{n}_t - \alpha^2 \hat{k}_{t-1} \hat{n}_t + \alpha \hat{k}_{t-1} \hat{z}_t - \alpha \hat{n}_t \hat{z}_t - \alpha^2 \hat{z}_t \hat{k}_{t-1} \hat{n}_t - \bar{N} (1-\alpha) \bar{z} \bar{K}^\alpha \bar{N}^{-\alpha} \cdot \hat{n}_t$$

$$\Rightarrow A \bar{C} + A \bar{C} \hat{c}_t = (1 - \bar{N}) (1-\alpha) \frac{\bar{Y}}{\bar{N}} + (1 - \bar{N}) (1-\alpha) \alpha \frac{\bar{Y}}{\bar{N}} \hat{k}_{t-1} + (1 - \bar{N}) (1-\alpha) \frac{\bar{Y}}{\bar{N}} \hat{z}_t - (1 - \bar{N}) (1-\alpha) \alpha \frac{\bar{Y}}{\bar{N}} \hat{n}_t - (1-\alpha) \bar{Y} \hat{n}_t$$

$$\Rightarrow A \bar{C} \hat{c}_t = (1-\alpha) \frac{1 - \bar{N}}{\bar{N}} \bar{Y} \left[ \alpha \hat{k}_{t-1} + \hat{z}_t - \alpha \hat{n}_t - \frac{\bar{N}}{1 - \bar{N}} \hat{n}_t \right]$$

$$\Rightarrow (1 - \bar{N}) (1-\alpha) \frac{\bar{Y}}{\bar{N}} \hat{c}_t = (1-\alpha) \frac{1 - \bar{N}}{\bar{N}} \bar{Y} \left[ \alpha \hat{k}_{t-1} + \hat{z}_t - \alpha \hat{n}_t - \frac{\bar{N}}{1 - \bar{N}} \hat{n}_t \right]$$

$$\Rightarrow \left( \alpha + \frac{\bar{N}}{1 - \bar{N}} \right) \hat{n}_t = \alpha \hat{k}_{t-1} + \hat{z}_t - \hat{c}_t$$

$$\Rightarrow \hat{n}_t = \left( \alpha + \frac{\bar{N}}{1 - \bar{N}} \right)^{-1} (\hat{z}_t + \alpha \hat{k}_{t-1} - \hat{c}_t)$$

$$(iv) 1 = \beta E_t \left[ \frac{C_t}{C_{t+1}} (\alpha z_{t+1} k_t^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta) \right]$$

$$\frac{C_t}{C_{t+1}} = \frac{\bar{C} e^{\hat{c}_t}}{\bar{C} e^{\hat{c}_{t+1}}} = e^{\hat{c}_t - \hat{c}_{t+1}} \approx 1 + \hat{c}_t - \hat{c}_{t+1} \Rightarrow \left( \frac{C_t}{C_{t+1}} \right) \approx \hat{c}_t - \hat{c}_{t+1}$$

$$M_{t+1} = \alpha z_{t+1} k_t^{\alpha-1} N_{t+1}^{1-\alpha} \Rightarrow \hat{m}_{t+1} = \hat{z}_{t+1} + (\alpha-1) \hat{k}_t + (1-\alpha) \hat{n}_{t+1}$$

$$\text{Let } G_{t+1} = M_{t+1} + 1 - \delta \Rightarrow \frac{G_{t+1} - \bar{G}}{\bar{G}} = \frac{M_{t+1} - \bar{M}}{\bar{G}} = \frac{\bar{M}}{\bar{G}} \frac{M_{t+1} - \bar{M}}{\bar{M}} \approx \frac{\bar{M}}{\bar{G}} \hat{m}_{t+1}$$



Let  $\tilde{\gamma} = \frac{\bar{M}}{\bar{G}} = \frac{\bar{G} - (1-\delta)}{\bar{G}} = 1 - \frac{1-\delta}{\bar{G}} \Rightarrow \tilde{\gamma} = 1 - \beta(1-\delta)$

$\bar{G} = \frac{1}{\beta}$

$\Rightarrow G_{t+1} \approx \bar{G} [1 + \tilde{\gamma} \hat{m}_{t+1}]$

$E_t \left[ \frac{C_t}{C_{t+1}} G_{t+1} \right] \approx E_t [1 + \hat{C}_t - \hat{C}_{t+1}] \bar{G} [1 + \tilde{\gamma} \hat{m}_{t+1}]$

Euler equation becomes:

$1 = \beta E_t [\bar{G} (1 + (\hat{C}_t - \hat{C}_{t+1}) + \tilde{\gamma} \hat{m}_{t+1})] = \beta \bar{G} (1 + E_t [\hat{C}_t - \hat{C}_{t+1} + \tilde{\gamma} \hat{m}_{t+1}])$

Since  $\beta \bar{G} = 1$

$\Rightarrow 1 = 1 + E_t [\hat{C}_t - \hat{C}_{t+1} + \tilde{\gamma} \hat{m}_{t+1}] \Rightarrow E_t [\hat{C}_t - \hat{C}_{t+1} + \tilde{\gamma} \hat{m}_{t+1}] = 0$

$\Rightarrow \hat{C}_t = E_t \hat{C}_{t+1} - \tilde{\gamma} E_t [\hat{z}_{t+1} - (1-\alpha) \hat{k}_t + (1-\alpha) \hat{n}_{t+1}]$

If you have CRRA,  $\frac{C_t}{C_{t+1}} \rightarrow \left( \frac{C_t}{C_{t+1}} \right)^\sigma \Rightarrow \sigma \hat{C}_t = \sigma E_t \hat{C}_{t+1} - E_t \hat{r}_{t+1}$

Thus, we have a system of equations:

$$\begin{cases} \hat{z}_t = \psi \hat{z}_{t-1} + \varepsilon_t \\ \hat{C}_t = E_t \hat{C}_{t+1} - \tilde{\gamma} E_t [\hat{z}_{t+1} - (1-\alpha) \hat{k}_t + (1-\alpha) \hat{n}_{t+1}] \\ \hat{n}_t = (\alpha + \frac{\bar{N}}{1-\bar{N}})^{-1} (\hat{z}_t + \alpha \hat{k}_{t+1} - \hat{C}_t) \\ \hat{k}_t = (\alpha \frac{\bar{Y}}{\bar{K}} + 1 - \delta) \hat{k}_{t-1} + (1-\alpha) \frac{\bar{Y}}{\bar{K}} \hat{n}_t + \frac{\bar{Y}}{\bar{K}} \hat{z}_t - \frac{\bar{C}}{\bar{K}} \hat{C}_t \end{cases}$$

Four equations, four variables:  $\hat{k}_{t+1}, \hat{C}_t, \hat{n}_t, \hat{z}_t$

$\hat{k}_{t+1}$  is predetermined variable,  $\hat{z}_t$  follows a stable AR(1) process.

$\hat{C}_t, \hat{n}_t$  are jump variables. Use undetermined coefficient method; guess

$\hat{k}_t = V_{kk} \hat{k}_{t-1} + V_{kz} \hat{z}_t, \hat{C}_t = V_{Ck} \hat{k}_{t-1} + V_{Cz} \hat{z}_t, \hat{n}_t = V_{nk} \hat{k}_{t-1} + V_{nz} \hat{z}_t$

$\Rightarrow \hat{k}_t = V_{kz} \sum_{i=0}^{\infty} \sum_{j=0}^i V_{kk}^j \psi^{i-j} \varepsilon_{t-i} \Rightarrow \hat{y}_t = V_{yz} \hat{C}_t + \sum_{i=0}^{\infty} \left( V_{yk} V_{kz} \sum_{j=0}^i V_{kk}^j \psi^{i-j} + V_{yz} \psi^{i+1} \right) \varepsilon_{t-i-1}$

$\hat{z}_t = \sum_{s=0}^{\infty} \psi^s \varepsilon_{t-s}$

$\hat{y}_t = \alpha \hat{k}_{t-1} + (1-\alpha) \hat{n}_t + \hat{z}_t$

$\Rightarrow \text{Var}(\hat{y}_t) = \int V_{yz}^2 + \sum_{i=0}^{\infty} \left[ V_{yk} V_{kz} \sum_{j=0}^i V_{kk}^j \psi^{i-j} + V_{yz} \psi^{i+1} \right]^2 \text{Var}(\varepsilon_t)$

Given parameters' value, we can back out the explicit numerical solution.

This model works: ① well for consumption volatility

② bad for output volatility. requires too small TFP volatility.



③ bad for hours volatility  $\rightarrow$  way too low

④ bad for investment volatility  $\rightarrow$  far too low.

But it allows  $N_t$  to move with shocks compared to Long-Plosser. While with this set of preference and technology, it cannot generate enough labor and investment volatility relative to output.

9 Hansen (1985) with indivisible labor.

Labor market phenomena:

① Hours fluctuate a lot relative to productivity.

② Micro evidence doesn't support huge intertemporal substitution of labor at individual level.

\* Target: Get big aggregate hours fluctuations without assuming each person is very willing to shift leisure across time.

Key mechanism: ① Each individual's labor is a corner-choice:  $\tilde{n}_t \in \{0, n^*\}$

Either work a fixed shift or no work.

② Lottery over jobs: assigned a job with probability  $\pi_t$ .

③ Complete unemployment insurance:  $C_t$  is independent of job status.

\* Crucial: we can write expected utility cleanly at the 'representative agent' level.

Thus, total labor supply:  $H_t = \bar{L} \pi_t n^*$

$\bar{L}$  is total labor force.

average labor supply:  $n_t = \frac{H_t}{\bar{L}} = \pi_t n^*$

Representative-agent utility: individual utility is  $U(C_t) + v(C_t \tilde{n}_t)$

Given lottery:  $E[U(C_t) + v(C_t \tilde{n}_t)] = U(C_t) + \pi_t v(C_t n^*) + (1 - \pi_t) v(0)$

Let  $v(0) = 0$  and  $\frac{v(C_t n^*)}{n^*} = -A < 0$   $\rightarrow C_t$  does not change with  $\tilde{n}_t$ , thus when take expectations, no  $\pi_t$  involved.

$$E[U(C_t) + v(C_t \tilde{n}_t)] = U(C_t) + \pi_t (-A n^*) = U(C_t) - A n_t$$

Suppose  $U(C_t) = \ln C_t \Rightarrow U_t = \ln C_t - A n_t$

$$\Rightarrow U_{Ct} = \frac{1}{C_t}, U_{nt} = -A$$

Micro-level: no smoothing in labor, either  $n^*$  or 0, keep small intertemporal labor substitution

Macro-level: Planner/Representative agent can vary  $\pi_t$  a lot, aggregate hours become very responsive.

Thus, we can define the equilibrium conditions for this model:



$$1 = \beta E_t \left[ \frac{C_t}{C_{t+1}} (\alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta) \right]$$

$$K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} + (1 - \delta) K_{t-1} - C_t$$

$$A C_t = (1 - \alpha) Z_t K_t^{\alpha} N_t^{1-\alpha} \rightarrow \text{key difference } - U_{N,t} = U_{C,t} W_t = U_{C,t} \text{MPL}_t$$

$(1 - N_t)$  term disappeared.

$$\log Z_t = (1 - \psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t$$

Mechanism: In indivisible labor model, movements in  $C_t$  translate into bigger movements in the intersection  $N_t$ . (Work more  $\rightarrow$  more people work (employment))  
 In divisible labor model,  $TFP \uparrow \Rightarrow$  Demand for labor  $\uparrow \Rightarrow W_t \uparrow \Rightarrow N_t \uparrow$   
 $1 - N_t \downarrow \Rightarrow \ln(1 - N_t)$  becomes more steep  $\Rightarrow$  the cost of giving up a marginal hour of leisure rises  
 $\Rightarrow$  Unwilling to adjust  $N_t$ , wage moves more  $\Rightarrow$  stiff labor  
 $\Rightarrow$  Small hour volatility. (every person chooses hours continuously)

Consistency with facts:

$$n_t = \frac{H_t}{L} = \frac{H_t}{L_t} \cdot \frac{L_t}{L} \approx n^* (1 - u_t) = n^* \pi_t$$

unemployment rate.

$$\frac{H_t}{L_t} = \bar{n}_t \approx n^* \text{ is empirically stable.}$$

\* Cyclical movement in total hours comes from (extensive margin) employment, not from hours per worker (intensive margin)

5) Other RBC models.

1° CRRA utility (King and Rebelo, 1999):  $U(C_t, L_t) = \ln C_t + A[(1 - N_t)^{1-\chi} - 1]/(1-\chi)$   
 \* Investment volatility  $\uparrow$  (almost correct), but underpredicts persistence, over-predicts comovement, consumption & labor are less volatile in model than data.

change preference curvature and more volatile TFP

2° KPR utility (King, Plosser and Rebelo, 1988):  $U(C_t, L_t) = \frac{[C_t r(L_t)]^{1-\frac{1}{\sigma}} - 1}{1-\frac{1}{\sigma}}$

\* Non-separable preference between  $C$  and  $N$ .  $U_{C,t}$  depends on  $L_t$ ,  $U_{L,t}$  depends on  $C_t$ .

Relax the too-tight linkages  $\Rightarrow$  improve comovement patterns.

3° GHH utility (Greenwood, Hercowitz-Huffman, 1988, AER)  $U(C_t, N_t) = \frac{1}{1-\frac{1}{\sigma}} \left[ \left( C_t - A \frac{N_t^{1+\psi}}{1+\psi} \right)^{1-\frac{1}{\sigma}} - 1 \right]$

\* Cancel wealth effect, labor supply responds to substitution effect (wage/productivity).  $\Rightarrow$  even more volatile  $N_t$

$$-U_{N,t} = U_{C,t} W_t \Rightarrow A N_t^{\psi} = W_t$$

4° Capital utilization:  $Y_t = Z_t (U_t K_{t+1})^{1-\alpha} N_t^{\alpha}$ . Extra intensive margin for capital.  
 $K_t = I_t + [1 - \delta u_t] K_{t-1} \Rightarrow$  higher investment volatility, higher output volatility.